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Progressions

A sequence is a list of #s. E.g. 1, 2, 3, 4, ... (sequence of natural #s).

Arithmetic progression (AP):
- The difference between any 2 consecutive terms is equal.
- E.g. 2, 5, 8, 11, 14.

$$a,\ a+d,\ a+2d,\ \ldots,\ \underbrace{a+(n-1)d}_{n^{th}\ term}$$

$$a_n = t_n = a+(n-1)d$$

Sum of n terms in AP:

$$S_n = \frac{n}{2}(a+l) = \frac{n}{2}(2a+(n-1)d)$$

Proof:

$$\begin{aligned} S_n &= a + (a+d) + (a+2d) + \cdots + (a+(n-1)d) \\ S_n &= (a+(n-1)d) + (a+(n-2)d) + \cdots + a \\ \hline 2S_n &= 2a+(n-1)d + 2a+(n-1)d + \cdots + 2a+(n-1)d \end{aligned}$$

$$2S_n = n[2a+(n-1)d]$$

$$S_n = \frac{n}{2}[2a+(n-1)d] = \frac{n}{2}[a+a+(n-1)d] = \frac{n}{2}[a+l]$$

$$\text{Average} = \frac{S_n}{n} = \frac{1}{2}(a+l)$$

Geometric progression (GP):
- The ratio of any 2 consecutive terms is equal.
- E.g. 2, 4, 8, 16, 32.

$$a,\ ar,\ ar^2,\ ar^3,\ \ldots,\ ar^{n-1}$$

$$a_n = ar^{n-1}$$

Sum:

$$\begin{aligned} S_n &= a + ar + ar^2 + \cdots + ar^{n-1} \\ r\cdot S_n &= ar + ar^2 + \cdots + ar^{n-1} + ar^n \\ \hline S_n(r-1) &= ar^n - a \end{aligned}$$

$$S_n = \frac{a(1-r^n)}{(1-r)} \qquad r \neq 1$$

Q. 2, 6, 18, 24, ... What is the sum of the first 6 terms?

→ $S_6 = \frac{2(3^6-1)}{(3-1)} = 3^6-1 = 728$

Q. 2, 4, 8, ..., 128. What is the # of terms?

→ $128 = 2\cdot 2^{n-1} = 2^n \Rightarrow n = 7$

Sum of infinite terms in GP:

If $|r|>1$ → $S_\infty = \infty$

If $|r|<1$ → $S_\infty = \frac{a}{1-r}$

$$\begin{aligned} S_\infty &= a + ar + ar^2 + \cdots \\ r\cdot S_\infty &= ar + ar^2 + \cdots \\ \hline (1-r)S_\infty &= a \end{aligned}$$

$$S_\infty = \frac{a}{(1-r)}$$

Harmonic progression (HP):

- Reciprocals of terms are in AP.

$2, 5, 8, 11, 14 \longrightarrow$ AP

$\frac{1}{2}, \frac{1}{5}, \frac{1}{8}, \frac{1}{11}, \frac{1}{14} \longrightarrow$ HP

$a_n = \frac{1}{a+(n-1)d}$

$S_n = \frac{1}{d} \ln\left(\frac{2a+(2n-1)d}{2n-d}\right)$ ↶ Sum of $\frac{1}{a}, \frac{1}{a+d}, \cdots, \frac{1}{a+(n-1)d}$

Arithmetic mean

$a_1, a_2, \ldots, a_n$

$AM = \frac{a_1+a_2+\cdots+a_n}{n}$

If a, b, c are in AP,

$2b = a+c$

Geometric mean

$a_1, a_2, \ldots, a_n$

$GM = \sqrt[n]{a_1 a_2 \cdots a_n}$

$= (a_1 \times a_2 \times \cdots \times a_n)^{1/n}$

for a, b, c in GP,

$b = \sqrt{ac}$

$GM = \sqrt[3]{abc}$

Harmonic mean

for $a_1, a_2, \ldots, a_n$

$HM = \frac{n}{\frac{1}{a_1}+\cdots+\frac{1}{a_n}}$

$GM^2 = AM \times HM$

$AM \geq GM \geq HM$] → for 2 given #s

NOTE:

- 3 #s in AP: $(a-d), a, (a+d)$
- 4 #s in AP: $(a-3d), (a-d), (a+d), (a+3d)$

If in an AP, sum of n terms is m and sum of m terms is n, then sum of m+n terms is -(m+n).

If in an AP, mth term = n, nth term = m, then the common difference is -1. The (m+n)th term = 0. The first term will be (m+n-1).

Proof:

$a+(m-1)d = n$

$a+(n-1)d = m$

$(m-n)d = (n-m)$

$d = -1$

$\Rightarrow a = m+n-1$

If the same number is added or subtracted from each term of an AP, the new series stays in AP with the same common difference.

If $a_1, a_2, \ldots, a_n$ is an AP, $d = a_2 - a_1$

$\Rightarrow a_1+k, a_2+k, \ldots, a_n+k$ is an AP. $d' = a_2+k-(a_1+k) = a_2-a_1 = d$ $\quad d' = d$.

If the same number is multiplied to every term of an AP, the new series stays in AP.

$a_1, a_2, \ldots, a_n \rightarrow$ AP

then $ka_1, ka_2, \ldots, ka_n \longrightarrow$ AP $\quad d' = ka_2 - ka_1 = k(a_2-a_1) = kd$ $\quad d' = kd$.

If the sum of m terms of an AP is equal to the sum of n terms of the same AP, then

$\left(\frac{m+n+1}{2}\right)^{th}$ term $= 0$ (if $m+n$ is odd)

if $m+n$ is even, it will not be 0.

Proof: $S_m = S_n$

$\Rightarrow \frac{m}{2}(2a+(m-1)d) = \frac{n}{2}(2a+(n-1)d)$

$\Rightarrow 2am + (m^2-m)d = 2an + (n^2-n)d$

$\Rightarrow 2a(m-n) + (m^2-m-n^2+n)d = 0$

$\Rightarrow a + \frac{(m^2-m-n^2+n)d}{2(m-n)} = 0$

$\Rightarrow a + \frac{[(m+n)(m-n) - 1(m-n)]d}{2(m-n)} = 0$

$\Rightarrow a + \frac{(m+n-1)d}{2} = 0$

$\Rightarrow a + \left(\frac{m+n+1}{2} - 1\right)d = 0$

$\Rightarrow a_{\frac{m+n+1}{2}} = 0$

Eg. $S_8 = S_{15}$ → $8+15 = 23$ → odd

$a_{\frac{8+15+1}{2}} = a_{12} = 0$

$S_8 = S_{15}$

$\Rightarrow \frac{8}{2}(2a+7d) = \frac{15}{2}(2a+14d)$

$\Rightarrow 16a + 56d = 30a + 210d$

$\Rightarrow 14a + 154d = 0$

$\Rightarrow a + 11d = 0$

$\Rightarrow a + (12-1)d = 0$

$\Rightarrow a_{12} = 0$

For natural #s, 1, 2, 3, ..., n, $1^3+2^3+3^3+\dots+n^3 = \left(\frac{n(n+2)}{2}\right)^2$

Q. $|a|<1$. Find $5+10a+15a^2+20a^3+\dots$

→ $S = 5+10a+15a^2+20a^3+\dots$

$aS = \quad 5a+10a^2+15a^3+\dots$

$S(1-a) = 5+5a+5a^2+5a^3+\dots$

$= 5(1+a+a^2+\dots)$

$S = \frac{5}{(1-a)^2}$

Q. The # of germs in a sample triples every 30 minutes. Find the # of germs in 180 minutes if 1000 was the initial # of germs.

→ 180 mins = 30×6

$a, 3a, 3^2a, 3^3a, 3^4a, 3^5a, 3^6a$

30 60 90 120 150 180

$ar^{6-1} = 1000(3)^6 = 729{,}000$

Practice questions

Q. $\frac{1}{2}+\frac{3}{4}+\frac{7}{8}+\cdots+\frac{4095}{4096}$

→ $S=\frac{1}{2}+\frac{3}{4}+\frac{7}{8}+\cdots+\frac{4095}{4096}$

$=\left(1-\frac{1}{2}\right)+\left(1-\frac{1}{4}\right)+\left(1-\frac{1}{8}\right)+\cdots+\left(1-\frac{1}{4096}\right)$

$=\left(1-\frac{1}{2^1}\right)+\left(1-\frac{1}{2^2}\right)+\left(1-\frac{1}{2^3}\right)+\cdots+\left(1-\frac{1}{2^{12}}\right)$

$=\underbrace{(1+1+1+\cdots+1)}_{12 \text{ terms}}+(-1)\left(\frac{1}{2}+\frac{1}{2^2}+\frac{1}{2^3}+\cdots+\frac{1}{2^{12}}\right)$

$=12-\frac{\frac{1}{2}\left(1-\frac{1}{2^{12}}\right)}{\left(1-\frac{1}{2}\right)}=12-1+\frac{1}{2^{12}}=\frac{11\cdot 2^{12}+1}{2^{12}}=\frac{45057}{4096}$

Q. In an AP, 10th term is 86, 86th term is 10, what'll be the 100th term?

→ $d=-1$ (if $a_n=m$ and $a_m=n$)

$a_{m+n}=0$

$a_{96}=0$

$a_{97}=-1$

$a_{98}=-2$

$a_{100}=-4$

Q. $S=\{2,3,4,\ldots,2n+1\}$.
X = avg of all odd integers of S.
Y = avg of all even integers of S. $X-Y=?$

→ $X=\frac{3+5+7+\cdots+(2n+1)}{n}=\frac{\frac{n}{2}(3+2n+1)}{n}=2+n$

$Y=\frac{2+4+\cdots+2n}{n}=\frac{\frac{n}{2}(2+2n)}{n}=n+1$

$X-Y=2+n-n-1=\underline{1}$

Q. # of terms from 30 to 530 divisible by 11 are ___.

→ $33, 44, \ldots, 528$

$11\cdot 3, 11\cdot 4, \ldots, 11\cdot 48$

$48-3+1=46$

Q. Split 74 into 4 parts s.t. all are in AP. Product of 1st & last is 2 less than that of 2nd and 3rd part.

→ $a-3d+a-d+a+d+a+3d=74$

$4a=74$

$a=18.5$

$a^2-9d^2=a^2-d^2-2$

$2=8d^2$

$d=\pm\frac{1}{2}$

#s are $18.5-1.5,\ 18.5-0.5,\ 18.5+0.5,\ 18.5+1.5$

$17, 18, 19, 20$

Q. A boy agrees to work for ₹1 on the 1st day, ₹2 on the 2nd, ₹4 on 3rd and so on.
How much will he get from 1st Feb to 20th Feb?

→ $1, 2, 4, \ldots$

$\frac{1(2^{20}-1)}{(2-1)}=2^{20}-1$

Q. 4th term of GP = 1/3. 10th term = 243. Find the 2nd term.

→ $\frac{ar^3}{ar^7} = \frac{1/3}{243}$ $\quad a \cdot 27 = \frac{1}{3}$ $\quad a = \frac{1}{81}$ $\quad a_2 = ar = \frac{3}{81} = \frac{1}{27}$

$r^6 = 729 = 3^6$

$r = 3$

Q. 7th term of GP is 8 times 4th term. 5th term is 48. 1st term = ?

→ $ar^6 = 8ar^3$ $\quad a_5 = ar^4 = 48$

$r^3 = 8 \Rightarrow r = 2$ $\quad \Rightarrow a = \frac{48}{16} = 3$

Q. Find k s.t. $\frac{2}{3}, k, \frac{5}{8}k$ are 3 consecutive terms in an AP.

→ $k - \frac{2}{3} = \frac{5}{8}k - k$

$\frac{3k-2}{3} = \frac{-3}{8}k$

$24k - 16 = -9k$

$33k = 16$

$k = 16/33$

Q. Find 9th term of HP, 6, 4, 3, ⋯

→ $\frac{1}{6}, \frac{1}{4}, \frac{1}{3}$ $\quad a_9 = \frac{1}{6} + 8 \cdot \frac{1}{12}$

$\frac{1}{4} - \frac{1}{6} = \frac{1}{12}$ $\quad \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$

$= \frac{1}{6} + \frac{4}{6} = \frac{5}{6}$ $\quad$ for HP → $\frac{6}{5}$.

Q. If $x > 1$, $y > 1$, $z > 1$ are in GP, then $\frac{1}{1+\log x}, \frac{1}{1+\log y}, \frac{1}{1+\log z}$ are in ____.

→ $\frac{y}{x} = \frac{z}{y}$

$y^2 = xz$

$2\log y = \log x + \log z$

$\log x, \log y, \log z \longrightarrow$ AP

$1+\log x, 1+\log y, 1+\log z \longrightarrow$ AP

$\frac{1}{1+\log x}, \frac{1}{1+\log y}, \frac{1}{1+\log z} \longrightarrow$ HP

Q. If a is the 1st term of GP, l is the nth term, p is the product of 1st n terms, p = ?

→ $l = ar^{n-1}$

$r = \left(\frac{l}{a}\right)^{\frac{1}{n-1}}$

$a \cdot ar \cdot ar^2 \cdot \cdots \cdot ar^{n-1}$

$a^n r^{\frac{(n-1)(n)}{2}}$

$= a^n \left(\frac{l}{a}\right)^{n/2}$

$= (al)^{n/2}$

Q. Find the sum of 1st 15 terms of $(2^2-1^2), (4^2-3^2), (6^2-5^2), \ldots$

→ $3, 7, 11, 15, \ldots$

$S_{15} = \frac{15}{2}[2\cdot 3 + 14\cdot 4] = 465$

Q. A certain bacteria quadruples every 45 mins. # of bacteria after 6 hrs if initially # of bact = 1500.

→ 6 hrs $= \frac{6 \times 60}{45} = 8$ 45 min periods.

After 8 hrs, $1500 \times 4^8 = 9.8304 \times 10^7$

Q. Find the sum of 1st 12 terms of seq. whose n^{th} term is $2n-10$.

→ $a_n = 2n-10$

$a = -8$

$a_2 = 4-10 = -6$ } seq. is AP

$a_3 = -4$

$d = 2$

$S_{12} = 6[2(-8) + 11\times 2] = 6[-16+22] = 36$

$a_n = 2n-10$

$= -10 + 2n$

$= -10 + (n-1)2 + 2$

$= (-8) + (n-1)(2)$

$= a \quad d$

Q. 3 integers form a GP. If 3rd term is decreased by 64, the #s will be in AP. If 2nd term of this AP is decreased by 8, GP again.

→

a, ar, ar^2

$a, ar, ar^2-64 \rightarrow$ AP

$ar - a = ar^2 - 64 - ar$

$a, ar-8, ar^2-64 \rightarrow$ GP

$(ar-8)^2 = a(ar^2-64)$

$a = 4, r = 5$ or $a = \frac{4}{9}, r = 13$ × we need int.

4, 20, 100

Q. If ratio of sum of x terms of 2 APs is $2x+3 : 3x+2$. Find ratio of nth terms.

→ $\frac{\frac{x}{2}[2a_1 + (x-1)d_1]}{\frac{x}{2}[2a_2 + (x-1)d_2]} = \frac{2x+3}{3x+2}$

$\frac{2a_1 + (x-1)d_1}{2a_2 + (x-1)d_2} = \frac{(2x+3)}{(3x+2)}$

Replace x by $2x-1$

$\frac{2a_1 + (2x-2)d_1}{2a_2 + (2x-2)d_2} = \frac{a_1 + (x-1)d_1}{a_2 + (x-1)d_2} = \frac{2(2x-1)+3}{3(2x-1)+2} = \frac{4x+1}{6x-1}$

Q. Sum of 3 #s in GP is 21. Sum of sqs = 189.

→ $a + ar + ar^2 = 21$

$a(1+r+r^2) = 21$

$a^2 + a^2r^2 + a^2r^4 = 189$

$a^2(1+r^2+r^4) = 189$

$\frac{1+r^2+r^4}{(1+r+r^2)^2} = \frac{189}{21\cdot 21} = \frac{3}{7}$

$7r^4 + 7r^2 + 7 = 3(1 + r^2 + r^4 + 2r^2 + 2r^6 + 2r^5)$

Let $r^2 = x$

solve $7x^2 + 7x + 7 = 3(1 + x + x^2 + 2x + 2x^3 + 2x^2)$

OR $\frac{(1+2r^2+r^4) - r^2}{(1+r+r^2)^2} = \frac{(1+r^2)^2 - r^2}{(1+r+r^2)^2} = \frac{(1+r+r^2)(1-r+r^2)}{(1+r+r^2)(1+r+r^2)} = \frac{3}{7}$

$$\frac{1-r+r^2}{1+r+r^2} = \frac{3}{7}$$

If $\frac{a}{b} = \frac{c}{d} \Rightarrow \frac{a+b}{a-b} = \frac{c+d}{c-d}$

$$\Rightarrow \frac{1-r+r^2+1+r+r^2}{1-r+r^2-1-r-r^2} = \frac{3+7}{3-7} = \frac{10}{-4} = -\frac{5}{2}$$

$$\Rightarrow \frac{2(1+r^2)}{-2r} = -\frac{5}{2}$$

$$\Rightarrow \frac{1+r^2}{r} = \frac{5}{2}$$

$$\Rightarrow 2r^2 - 5r + 2 = 0$$

$$\Rightarrow 2r^2 - 4r - r + 2 = 0$$

$$\Rightarrow 2r(r-2) - (r-2) = 0$$

$$\Rightarrow r = 2, \frac{1}{2}$$

$r = 2$

$a(1+r+r^2) = 21$

$a(1+2+4) = 21$

$a = 3$

$r = 1/2$

$a\left(1+\frac{1}{2}+\frac{1}{4}\right) = 21$

$a\left(\frac{4+2+1}{4}\right) = 21$

$a = 12$

Q. $\frac{3}{4} - \frac{5}{4^2} + \frac{3}{4^3} - \frac{5}{4^4} + \frac{3}{4^5} - \cdots$

$\rightarrow 3\left(\frac{1}{4} + \frac{1}{4^3} + \frac{1}{4^5} + \frac{1}{4^7} + \cdots\right) - 5\left(\frac{1}{4^2} + \frac{1}{4^4} + \frac{1}{4^6} + \cdots\right)$

$= 3 \cdot \frac{1/4}{1-(1/4)^2} - 5 \cdot \frac{1/4^2}{1-\frac{1}{4^2}}$

$= 3 \cdot \frac{1}{4} \cdot \frac{16}{15} - 5 \cdot \frac{1}{16} \cdot \frac{16}{15}$

$= \frac{12}{15} - \frac{5}{15} = \frac{7}{15}$